

Research Journal

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September

Sep. 16, 2025 — Tuesday

- Revisited the credentials provided by the HPC facility at IUCAA last year – they are working fine
- Here are the current credentials:
 - **VPN**
 - * Server: moon.iucaa.in:4433
 - * Username: parag.bhattacharya
 - * Password: P@^@GBvpn2*2486
 - * Domain: LocalDomain
 - **ASSC account**
 - * Command: ssh -X paragb@192.168.11.34
 - * Password: same as Win11
 - **Using the installed software**

```
source xraysoftinit
heainit
sas21init
sas21
```
 - **Using Python3.10 etc.**

```
module switch python/python310
```
 - **LDAP account credentials (at IUCAA)**
 - * Username: parag
 - * Password: Saturn\$890

Sep. 17, 2025 — Wednesday

- Hopefully I will be able to do some polarimetric study of science data from IXPE, as proposed by Dr. Monmoyuri Baruah yesterday.
- The target for doing some work is 6 – 8 October 2025, during NEMA-XI at RGU, Arunachal Pradesh.
- I have two documents that I need to study:
 1. [Bobrikova et al. \(2024\)](#) – sent by Dr. Monmoyuri Baruah
 2. [IXPE Quick Start Data Analysis Guide](#) – downloaded from HEASARC
- To get started with the data analysis, I want to download the IXPE L2 data for two sources:
 1. GX 13+1 – a weakly magnetized neutron star (WMNS) studied by [Bobrikova et al.](#)
 2. Mrk 501 – a blazar which is provided as an example in the [IXPE Quick Start Guide](#)
- The science data are to be downloaded from the [legacy HEASARC browse window](#) as per the following parameters:
 - Object Name or Coordinates: Mrk 501
Mission: IXPE
Observation window: July 2022
 - Object Name or Coordinates: GX 13+1
Mission: IXPE
Observation window: October 2023

Note: The current Xamin browse portal doesn't seem to load

Sep. 21, 2025 — Sunday

- Downloaded the L2 event files for the following dataset:
 1. Mrk 501
 2. GX 13+1
- Commands for file transfer using SCP – from [Contabo Blog](#)
- Commands for Mrk 501
 - Uploaded Mrk 501 .tar file from PC to ASSC server:

```
scp E:\Research\x-ray_polarimetry\data\raw\mrk-501.tar  
paragb@192.168.11.34:/home/paragb/xray-pol/dataset
```

- Navigated to data folder:
`cd /home/paragb/xray-pol/dataset`
- Extracted data from the .tar file
`tar -xvf mrk-501.tar`
- The extracted L2 files are contained in the folder:
`/home/paragb/xray-pol/dataset/01004701/event_l2`
- The data folder has been accordingly renamed as:
`mv -v 01004701/ mrk-501/`
- So the extracted L2 files are now contained in the folder:
`/home/paragb/xray-pol/dataset/mrk-501/event_l2`
- The .gz files are unzipped as:
`gunzip -v *.gz`

Sep. 22, 2025 — Monday

- Downloaded the relevant CALDB files for IXPE to the ASSC server using the commands

```
export CALDB=/home/paragb/caldb/  
cd $CALDB  
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/  
ixpe/gpd/goodfiles_ixpe_gpd_20250225.tar.gz  
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/  
ixpe/xrt/goodfiles_ixpe_xrt_20241028.tar.gz  
gunzip -v goodfiles_*.gz  
tar -xvf goodfiles_*.tar  
rm -rv goodfiles_*.tar
```

- Downloaded the data .tar files for Mrk 501 directly to the ASSC server.

Sep. 23, 2025 — Tuesday

- Began download of Heasoft v6.35.2 from remote server using the following command after opening the terminal in the desired folder.

```
wget -v https://heasarc.gsfc.nasa.gov/FTP/  
software/lheasoft/lheasoft6.35.2/heasoft-6.35.2src.tar.gz
```

- Installed MiKTeX using the following commands ([MiKTeX webpage](#)):

```
curl -fsSL https://miktex.org/download/key |  
    sudo gpg --dearmor -o /usr/share/keyrings/miktex.gpg  
echo "deb [signed-by=/usr/share/keyrings/miktex.gpg]  
    https://miktex.org/download/ubuntu jammy universe" |  
    sudo tee /etc/apt/sources.list.d/miktex.list  
sudo apt-get update  
sudo apt-get install miktex  
miktexsetup finish  
initexmf --set-config-value [MPM]AutoInstall=1
```

- Installed TeXstudio using the following commands:

```
sudo add-apt-repository ppa:sunderme/texstudio  
sudo apt update  
sudo apt install texstudio
```

- The .tar data files have been downloaded into the ASSC server at

/home/paragb/xray-pol/dataset/mrk-501/01004701/

- All .gz files have been extracted in the following folders:

auxil event_l1 event_l2 hk

- To view the images on DS9: `ds9 ixpe01004701_det1_evt2_v01.fits &`

- Enhancement of the image:
scale → log
color → heat
- Creation of a source region:
In the menu: e.g. Region → Shape → Circle
edit → region → click + drag to draw the region
Double-click on region to open the region dialog. Center in fk5, radius in arcsec.
- Creation of a background region:
In the menu: e.g. Region → Shape → Annulus
edit → region → click + drag to draw the region
Double-click on region to open the region dialog. Center in fk5, radius in arcsec.

- The region files are created in the L2 event folder. These files contain the following:

- Inside `src.reg`:

```
circle(16:53:51.766,+39:45:44.41,60.0)
```

- Inside `bkg.reg`:

```
circle(16:53:51.766,+39:45:44.41,252.0)
-circle(16:53:51.766,+39:45:44.41,132.0)
```

- Extraction of spectra

- Initializing `xselect`

```
xselect prefix=mrk501
xsel> read event "ixpe01004701_det1_evt2_v01.fits"
```

- Extracting the source spectra (e.g. detector 1):

```
xsel> filter region "src.reg"
xsel> extract SPEC stokes=NEFF
xsel> save spec mrk-501_d01_src_
```

This creates 3 source spectra files, namely

```
mrk-501_d01_src_I.pha
mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
```

- Extracting the background spectra (e.g. detector 1):

```
xsel> clear region
xsel> filter region "bkg.reg"
xsel> extract SPEC stokes=NEFF
xsel> save spec mrk-501_d01_bkg_
```

This creates 3 background spectra files, namely

```
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
```

- Finding the appropriate RMF files:

1. Find the observation in the `ixmaster` catalogue using the [HEASARC search form](#)
2. Copy the entry for the object of interest under the `time` column, e.g. 2022-07-09 23:23:24.082 for the Mrk 501 obs. ID 01004701

3. Create a CALDB variable that points to the folder containing the IXPE calibration database as

```
export CALDB=/home/paragb/caldb
```

4. Use the time value in the quzCIF command, for detector unit 1 (DU1) of the gas pixel detector instrument (GPD), as follows:

```
quzCIF IXPE GPD DU1 - MATRIX 2022-07-09 23:23:24.082 -
```

This would give the following output:

```
/home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_01.rmf 1
/home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_alpha075_01.rmf 1
```

5. We have to use the file whose name contains the string alpha075 when the events are weighted by either the NEFF or SIMPLE weighting schemes. In our case, we used NEFF while extracting the source and background spectra and so we can create a soft link to the alpha075 file as follows:

```
ln -s /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_alpha075_01.rmf mrk-501_d01_rmf.fits
```

This creates a symbolic link file named mrk-501_d01_rmf.fits which can be used as the RMF file in XSPEC later.

- Creating the MRF (for Q and U spectra) and ARF (for I spectrum) files:

1. Sent the following error report to NASA Helpdesk:

Recently I have been introduced to IXPE science data analysis and I have been trying to replicate the results of the walkthrough in section 5 of the IXPE quick start analysis guide.

I have been able to successfully generate the I, Q and U spectra as described in the aforementioned document. However, I am running into an error when I try to create the MRF file for the Q and U spectra.

I have copied the attitude file from the /hk folder into the /event_l2 folder. The input and output of the console are as follows:

```
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
```

```
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
[ixpe01004701_det1_att_v01.fits]
attfiles=ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'
```

The issue persists even when I provide the full path name of the attitude file:

```
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=/home/paragb/xray-pol/dataset/mrk-501/
01004701/event_l2/ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
[attfiles=ixpe01004701_det1_att_v01.fits.gz]
/home/paragb/xray-pol/dataset/mrk-501/01004701/event_l2/
ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'
```

I am bamboozled because I have followed the exact same steps as given in the document. I am unable to figure out as to where I have been going wrong.

I would be grateful to receive advice from you in this matter.

Sep. 24, 2025 — Wednesday

- Received the following response from NASA Helpdesk:

Hello,

We were unable to reproduce the error you reported.
To help resolve this issue, please try the following:

1. Execute 'punlearn ixpecalcarf' to clean your parameter file, as we suspect there may be a stale parameter value causing the problem.
2. Please also check your attitude file configuration, as your output shows the attitude file name appearing inconsistently with and without the .gz extension.

Please let us know if this resolves the issue or if you continue to experience problems.

Thanks,
Mihoko Yukita

- Installed Heasoft v6.35.2 on laptop.

Sep. 25, 2025 — Thursday

- Started analysis of data for Mrk 501 locally on my laptop.
- Installed Heasoft 6.35.2 locally on my PC (running on Ubuntu 22.04 LTS).
- Remote access for CALDB set up in the folder:
/home/pararover/Software/astro-soft/caldb
which contains the files alias_config.fits and caldb.config
- The .bashrc file is appended with the following lines:

```
CALDBCONFIG=/home/pararover/Software/astro-soft/caldb/
caldb.config; export CALDBCONFIG
CALDBALIAS=/home/pararover/Software/astro-soft/caldb/
alias_config.fits; export CALDBALIAS
CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb; export CALDB
```

- Data from HEASARC are downloaded and extracted into the following folder:

```
/home/pararover/Documents/Research/xray-pol/dataset/mrk-501/01004701
```

Sep. 28, 2025 — Sunday

- Downloaded .tar files for IXPE observations GX 13+1, with the Obs. ID. 02006801, in two batches because HEASARC does not allow downloads more than 2 GB.
 1. event_l1/ for ~ 1.6 GB
 2. event_l2/, auxil/ and hk/ for ~ 700 MB
- The above two batches of .tar files need to be extracted and combined together into a single folder named gx13+1/ with the following folder hierarchy:
 - gx13+1/
 - * 02006801

- auxil/
- event_l1/
- event_l2/
- hk/

October

Oct. 01, 2025 — Wednesday

- Found this good paper on polarized X-ray emission from a BH XRB named 4U 1630-47

<https://iopscience.iop.org/article/10.3847/2041-8213/acd77b>

- For their analysis Rawat et al. have used IXPEOBSSIM

<https://ixpeobssim.readthedocs.io/en/latest/index.html>

- I intend to explore IXPEOBSSIM further in this direction of research. To this end, I have installed the same on my PC using the command (after running heainit):

```
pip3 install ixpeobssim
```

- IXPEOBSSIM requires the following Python packages to run:

```
numpy    scipy    matplotlib    astropy    regions    skyfield
```

- I have upgrade all of the above packages individually by running the command:

```
pip3 install <<package-name>> --upgrade
```

- Before proceeding, I need to look into the tutorials provided here:

<https://ixpeobssim.readthedocs.io/en/latest/tutorial.html>

- Logs from my tryout of the tutorial:

```
>>> import numpy
>>>
>>> from ixpeobssim.config import file_path_to_model_name
>>> About to create folder /home/pararover/ixpeobssimdata...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimdata/benchmarks...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimauxfiles...
>>> Folder succesfully created.
Matplotlib is building the font cache; this may take a moment.
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/__init__.py:63:
UserWarning: Unable to import Axes3D. This may be due to
multiple versions of Matplotlib being installed
```

```

    (e.g. as a system package and as a pip package). As a result,
    the 3D projection is not available.
    warnings.warn("Unable to import Axes3D.
    This may be due to multiple versions of "
>>> from ixpeobssim.srcmodel.roi import xPointSource, xROIModel
>>> from ixpeobssim.srcmodel.spectrum import power_law
>>> from ixpeobssim.srcmodel.polarization import constant
>>>

```

Oct. 16, 2025 — Thursday

- The solution provided by the IXPE Helpdesk is working.
- Was able to run `ixpecalcarf` to generate MRF files for D01, D02 and D03.
- However, the following warning appeared:

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02.mrf \
> specfile=None radius=1.0 weight=1 resptype=mrf
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/_init_.py:63: UserWarning: Unable to import Axes3D.
This may be due to multiple versions of Matplotlib being
installed (e.g. as a system package and as a pip package). As a
result, the 3D projection is not available.
    warnings.warn("Unable to import Axes3D. This may be due to
    multiple versions of "

```

- The `ixpecalcarf` task has been run as follows:

1. For detector #1:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_QU.mrf \
> specfile=None radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

2. For detector #2:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

3. For detector #3:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

- The list of necessary files for detector #1 are:

mrk-501_d01_src_I.pha

```
mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
mrk-501_d01.rmfm -> ixpe_d1_20211209_alpha075_01.rmfm
mrk-501_d01_I.arfm
mrk-501_d01_QU.rmfm
```

- The list of necessary files for detector #2 are:

```
mrk-501_d02_src_I.pha
mrk-501_d02_src_Q.pha
mrk-501_d02_src_U.pha
mrk-501_d02_bkg_I.pha
mrk-501_d02_bkg_Q.pha
mrk-501_d02_bkg_U.pha
mrk-501_d02.rmfm -> ixpe_d2_20211209_alpha075_01.rmfm
mrk-501_d02_I.arfm
mrk-501_d02_QU.rmfm
```

- The list of necessary files for detector #3 are:

```
mrk-501_d03_src_I.pha
mrk-501_d03_src_Q.pha
mrk-501_d03_src_U.pha
mrk-501_d03_bkg_I.pha
mrk-501_d03_bkg_Q.pha
mrk-501_d03_bkg_U.pha
mrk-501_d03.rmfm -> ixpe_d2_20211209_alpha075_01.rmfm
mrk-501_d03_I.arfm
mrk-501_d03_QU.rmfm
```

- Commands for spectrum file grouping of detector #1:

1. For *I* spectrum:

```
fthedit mrk-501_d01_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_I.pha' " longstring=YES
```

```
fthedit mrk-501_d01_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmfm' " longstring=YES
```

```
fthedit mrk-501_d01_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d01_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d01_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmrf'" longstring=YES
```

```
fthedit mrk-501_d01_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.rmrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d01_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d01_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmrf'" longstring=YES
```

```
fthedit mrk-501_d01_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.rmrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #2:

1. For I spectrum:

```
fthedit mrk-501_d02_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_I.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmrf'" longstring=YES
```

```
fthedit mrk-501_d02_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d02_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmrf'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d02_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmrf'" longstring=YES
```

```
fthedit mrk-501_d02_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #3:

1. For I spectrum:

```
fthedit mrk-501_d03_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_I.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmrf'" longstring=YES
```

```
fthedit mrk-501_d03_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d03_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmrf'" longstring=YES
```

```
fthedit mrk-501_d03_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

3. For U spectrum:

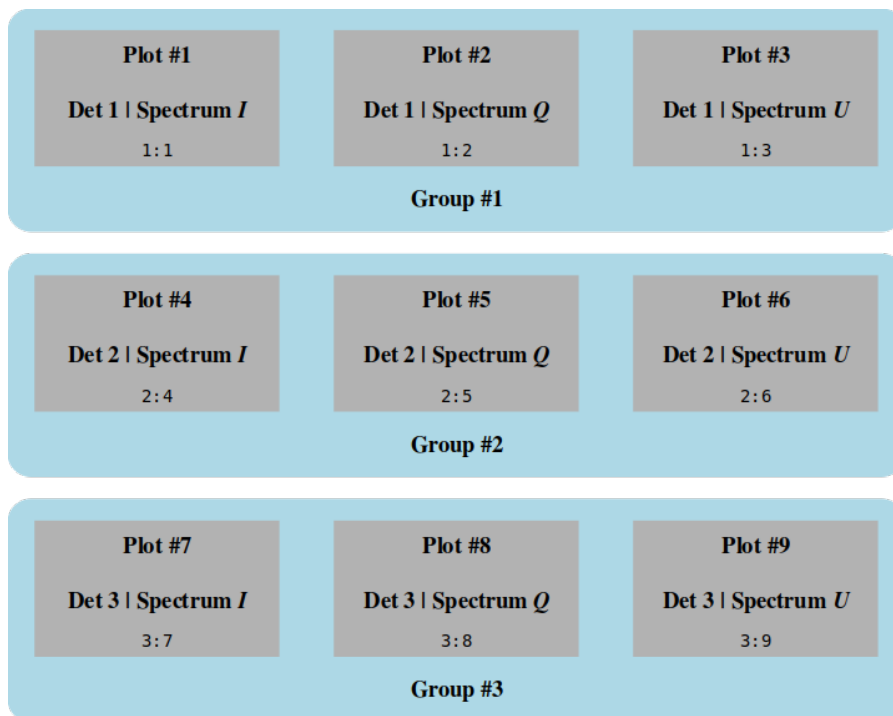
```
fthedit mrk-501_d03_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmrf'" longstring=YES
```

```
fthedit mrk-501_d03_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

Oct. 17, 2025 — Friday

- Spectral fitting of all data groups to be performed as per the Quick Start Manual.
- Understanding the data and plot groups in XSPEC:



- The above data groups are created in XSPEC as follows:

```
data 1:1 mrk-501_d01_src_I.pha
data 1:2 mrk-501_d01_src_Q.pha
data 1:3 mrk-501_d01_src_U.pha

data 2:4 mrk-501_d02_src_I.pha
```



```
data 2:5 mrk-501_d02_src_Q.pha
data 2:6 mrk-501_d02_src_U.pha
```

```
data 3:7 mrk-501_d03_src_I.pha
data 3:8 mrk-501_d03_src_Q.pha
data 3:9 mrk-501_d03_src_U.pha
```

Oct. 18, 2025 — Saturday

- Steps that were used to analyze the spectra of Mrk 501 in XSPEC are given below:

I. Reading the data:

```
XSPEC12> data 1:1 mrk-501_d01_src_I.pha
XSPEC12> data 1:2 mrk-501_d01_src_Q.pha
XSPEC12> data 1:3 mrk-501_d01_src_U.pha
XSPEC12> data 2:4 mrk-501_d02_src_I.pha
XSPEC12> data 2:5 mrk-501_d02_src_Q.pha
XSPEC12> data 2:6 mrk-501_d02_src_U.pha
XSPEC12> data 3:7 mrk-501_d03_src_I.pha
XSPEC12> data 3:8 mrk-501_d03_src_Q.pha
XSPEC12> data 3:9 mrk-501_d03_src_U.pha
```

II. Filtering the photons within an energy bandwidth:

```
XSPEC12> ignore *:-2.0 8.0-*
```

Here the first * means to apply this filter on all data groups. Then *-2.0 8.0-* means to ignore all photon counts for $E < 2 \text{ keV}$ and $E > 8 \text{ keV}$, i.e. only the $2 \text{ keV} \leq E \leq 8 \text{ keV}$ bandwidth is to be considered.

III. Applying the composite model:

```
XSPEC12> model constant*tbabs*(polconst*powerlaw)
```

Here constant models an energy-independent multiplicative factor, tbabs models the intervening absorption, polconst models a constant polarization at all energies and powerlaw models a simple photon power law.

IV. Parameter #1 is frozen at 1. Parameters 7 and 13 are untied and thawed. The parameter list prior to fitting looks as follows:

```
XSPEC12> show par
```

Parameters defined:

```

=====
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component Parameter Unit Value
par comp
Data group: 1
1 1 constant factor 1.000000 frozen
2 2 TBabs nH 10^22 1.000000 +/- 0.0
3 3 polconst A 1.000000 +/- 0.0
4 3 polconst psi deg 45.0000 +/- 0.0
5 4 powerlaw PhoIndex 1.000000 +/- 0.0
6 4 powerlaw norm 1.000000 +/- 0.0
Data group: 2
7 1 constant factor 1.000000 +/- 0.0
8 2 TBabs nH 10^22 1.000000 = p2
9 3 polconst A 1.000000 = p3
10 3 polconst psi deg 45.0000 = p4
11 4 powerlaw PhoIndex 1.000000 = p5
12 4 powerlaw norm 1.000000 = p6
Data group: 3
13 1 constant factor 1.000000 +/- 0.0
14 2 TBabs nH 10^22 1.000000 = p2
15 3 polconst A 1.000000 = p3
16 3 polconst psi deg 45.0000 = p4
17 4 powerlaw PhoIndex 1.000000 = p5
18 4 powerlaw norm 1.000000 = p6
=====

```

V. Renormalizing and fitting the model to the spectrum gives the following parameter values:

```

=====
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component Parameter Unit Value
par comp
Data group: 1
1 1 constant factor 1.000000 frozen
2 2 TBabs nH 10^22 0.541250 +/- 0.117460
3 3 polconst A 7.25453E-02 +/- 1.90183E-02
4 3 polconst psi deg -49.1296 +/- 7.50921
5 4 powerlaw PhoIndex 2.60742 +/- 5.68093E-02
6 4 powerlaw norm 9.63365E-02 +/- 8.13714E-03
Data group: 2

```

7	1	constant	factor		1.00375	+/-	9.94110E-03
8	2	TBabs	nH	10 ²²	0.541250	=	p2
9	3	polconst	A		7.25453E-02	=	p3
10	3	polconst	psi	deg	-49.1296	=	p4
11	4	powerlaw	PhoIndex		2.60742	=	p5
12	4	powerlaw	norm		9.62997E-02	=	p6
Data group: 3							
13	1	constant	factor		0.924722	+/-	1.00195E-02
14	2	TBabs	nH	10 ²²	0.541250	=	p2
15	3	polconst	A		7.25453E-02	=	p3
16	3	polconst	psi	deg	-49.1296	=	p4
17	4	powerlaw	PhoIndex		2.60742	=	p5
18	4	powerlaw	norm		9.63365E-02	=	p6

Fit statistic: Chi-Squared 131.57 using 149 bins,
spectrum 1, group 1.

Chi-Squared	159.05	using 149 bins, spectrum 2, group 1.
Chi-Squared	122.69	using 149 bins, spectrum 3, group 1.
Chi-Squared	130.71	using 149 bins, spectrum 4, group 2.
Chi-Squared	118.58	using 149 bins, spectrum 5, group 2.
Chi-Squared	157.11	using 149 bins, spectrum 6, group 2.
Chi-Squared	60.87	using 149 bins, spectrum 7, group 3.
Chi-Squared	82.38	using 149 bins, spectrum 8, group 3.
Chi-Squared	69.58	using 149 bins, spectrum 9, group 3.
Total fit statistic	1032.53	with 1334 d.o.f.

Test statistic : Chi-Squared 1032.53 using 1341 bins.
Null hypothesis probability of 1.00e+00 with 1334 degrees of freedom

VI. We are interested in finding the errors in the polarization degree A and polarization angle ψ , which are represented by the parameters A and ψ in the `polconst` model component.

- To find the 99% confidence intervals for, say data group 1, we have

```
XSPEC12> error 6.635 3
```

```
XSPEC12> error 6.635 4
```

- To find the 1σ confidence intervals for, say data group 1, we have

```
XSPEC12> error 1.0 3
```

```
XSPEC12> error 1.0 4
```

Note: We cannot compute the errors in the same parameters for the other two data groups (i.e. 2 and 3) because these parameters are tied to their counterparts in data group 1. Any attempt to do so will be flagged as an error by XSPEC as follows:

```
XSPEC12> error 6.635 9
*** Parameter 9 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 10
*** Parameter 10 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 15
*** Parameter 15 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 16
*** Parameter 16 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
```

VII. To visualize the contour plot, we need the following commands:

```
XSPEC12> steppar 3 0.00 0.21 50 4 -90 0 60
XSPEC12> plot contour , , 4 1.386, 4.61 9.21 13.81
```

The steppar command generates a 2D grid of A and ψ values with 50 steps in the range $0 < A < 0.21$ and 60 steps in the range $-90^\circ < \psi < 0^\circ$.

The plot contour command plots contours at 50% (1.386), 90% (4.61), 99% (9.21) and 99.9% (13.81) confidence levels for 2D errors in χ^2 .

- To estimate the MDP_{99} using PIMMS, we need to do the following:

1. Determine the model flux in the $2\text{ keV} \leq E \leq 8\text{ keV}$ bandwidth:

```
XSPEC12> flux 2.0 8.0
Spectrum Number: 1, 2, 3
Data Group Number: 1
Model Flux 0.015886 photons (8.7966e-11 ergs/cm^2/s) range
(2.0000 - 8.0000 keV)

Spectrum Number: 4, 5, 6
Data Group Number: 2
Model Flux 0.015945 photons (8.8297e-11 ergs/cm^2/s) range
```

(2.0000 - 8.0000 keV)

Spectrum Number: 7, 8, 9

Data Group Number: 3

Model Flux 0.01469 photons (8.1345e-11 ergs/cm²/s) range
(2.0000 - 8.0000 keV)

2. Collect the following values from the best-fit parameters so far:

$n_H = 0.540529 \times 10^{22} \text{ cm}^{-2}$ (Galactic H column density)

$\Gamma = 2.60715$ (photon index from powerlaw)

$\langle F \rangle = \left(\frac{8.7966 + 8.8297 + 8.1345}{3} \right) 10^{-11} = 8.5869 \times 10^{-11}$ (avg. flux from 3 detectors)

3. Enter these values, along with some more, in **WebPIMMS** as shown:



A Mission Count Rate Simulator
Powered by [PIMMS v4.15](#)

Access the [multiple component model interface](#).

Convert From:		Into:	
Flux		IXPE	
Examples of Common FLUX Input/Output Ranges			
Input Energy Range (low-high): 2-8	☒ keV ☐ Angstroms		Units
Output Energy Range (low-high): 2-8	☒ keV ☐ Angstroms		Units

Source Flux / Count Rate: 8.5869e-11		(erg/cm ² /s) (counts/s)
Galactic nH	Redshift	Intrinsic nH
0.540529e22 (cm ⁻²)	none	none (cm ⁻²)

Model of Source:	Model Parameters
☒ Power Law	Photon Index: 2.60715
☐ Black Body	keV:
☐ Therm. Bremss.	kT:
☐ APEC	1.0 Solar Abundance LogT keV

4. **WebPIMMS** computes and displays the predicted count rates and MDP_{99} values using different algorithms, some of which are shown below: We find,

IXPE OPEN Modulation Factor
Modulation Factor = 0.3140 In 10000.0 s, MDP_{99} = 16.13% In 100000.0 s, MDP_{99} = 5.10%
IXPE GRAY TR21D
The Total count rate in the selected band: PIMMS predicts 1.817E-01 cps with IXPE GRAY TR21D
IXPE GRAY ER21D
The Effective count rate in the selected band: PIMMS predicts 1.549E-01 cps with IXPE GRAY ER21D
IXPE GRAY MR21D
The Modulated count rate in the selected band: PIMMS predicts 6.300E-02 cps with IXPE GRAY MR21D
IXPE GRAY Modulation Factor
Modulation Factor = 0.4069 In 10000.0 s, MDP_{99} = 26.80% In 100000.0 s, MDP_{99} = 8.48%

for instance, under ‘IXPE OPEN Modulation Factor’, that for an exposure time of 100 ks, the MDP_{99} is 5.15%.

$$T = 100 \text{ ks} \implies MDP_{99} = 5.10\% = 0.051$$

Now MDP_{99} scales with exposure time T as $MDP_{99} \propto \frac{1}{\sqrt{T}}$. So we have

$$\frac{MDP_{99}(\text{actual})}{MDP_{99}(\text{simulated})} = \sqrt{\frac{T(\text{simulated})}{T(\text{actual})}}$$

Running the show data command gives us the actual exposure time (e.g. for I spectrum by detector #1) to be 97800 s = 97 ks.

Therefore, substituting $T(\text{simulated}) = 100 \text{ ks}$, $T(\text{actual}) = 97 \text{ ks}$ and $MDP_{99}(\text{simulated}) = 5.10\%$ above, we obtain

$$MDP_{99}(\text{actual}) = \sqrt{\frac{100}{97}} (5.10\%) = 5.18\% = 0.0518$$

This implies that, for an *unpolarized source* with the same physical parameters, an IXPE observation will return a value $A > 0.0518$ only 1% (=100%-99%, the 99 is from MDP_{99}) of the time.

- The chain of logic is as follows:

1. Scale PIMMS number because the real observation time was a bit shorter, i.e. $MDP_{99} = 5.18\%$.
2. MDP_{99} means “only 1% chance a zero-polarisation source would look this big or bigger.”
3. Measured value of 7.24% (from best-fit) is above 5.18%, and the measurement’s uncertainty of 1.9% is small (i.e. $5.18/1.9 \approx 3.8$).
4. Therefore the measured polarisation is very unlikely to be a random fluke – it is a highly probable real detection.

2026

May

May. 17, 2026 — Sunday

- All data have been now migrated to the following location on my PC:

`/home/pararover/AstroDA/xray-pol/dataset`

May. 27, 2026 — Wednesday

- Checked my VPN and ASSC account credentials at the HPC facility in IUCAA which have been lying dormant for quite some time – both are working fine.
- For GX13+1 I had downloaded the data corresponding to the ObsID 02006801.
- Extracted from the .tar file to find the following folder hierarchy:
 - 02006801
 - * auxil/
 - * event_l1/
 - * event_l2/
 - * hk/
- We need to gunzip all the files in each of the sub-folders above.
- In addition, we need to copy the following files from the hk/ folder and paste into the event_l2/ folder:
 - ixpe02006801_det1_att_v01.fits
 - ixpe02006801_det2_att_v01.fits
 - ixpe02006801_det3_att_v01.fits
- As per section 2.1 in [Bobrikova et al. \(2024\)](#), the “source photons were selected in a circular region with a radius of 100" centered at the source position.”

- To obtain the source region, we need to open any of the L2 event files using DS9. For this, we have to navigate to the event_l2/ folder and open any of the files from detector 1, 2 or 3 using DS9, i.e.

```
ds9 ixpe02006801_det1_evt2_v01.fits &
```

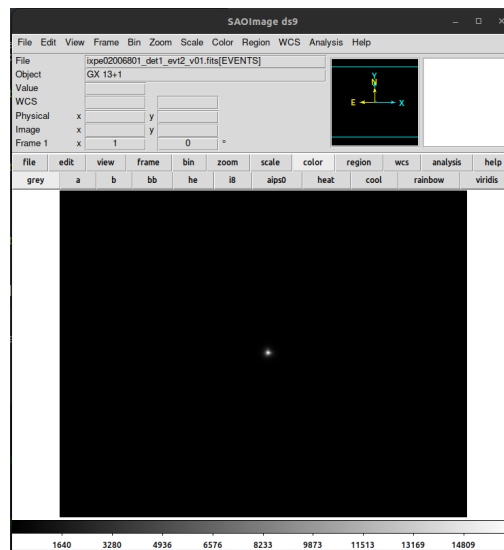
or

```
ds9 ixpe02006801_det2_evt2_v01.fits &
```

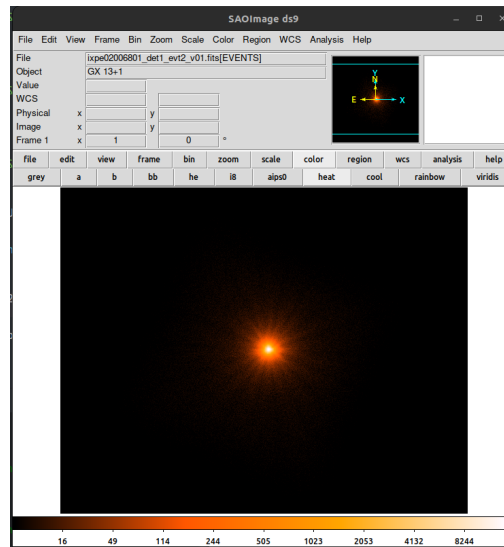
or

```
ds9 ixpe02006801_det3_evt2_v01.fits &
```

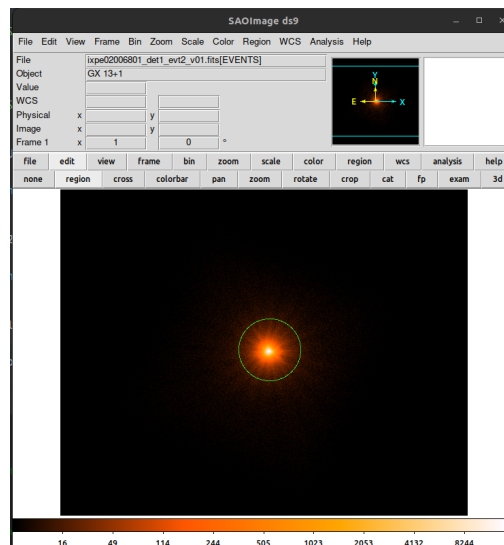
- Suppose we choose the file ixpe02006801_det1_evt2_v01.fits. Upon opening in DS9, we obtain as follows:



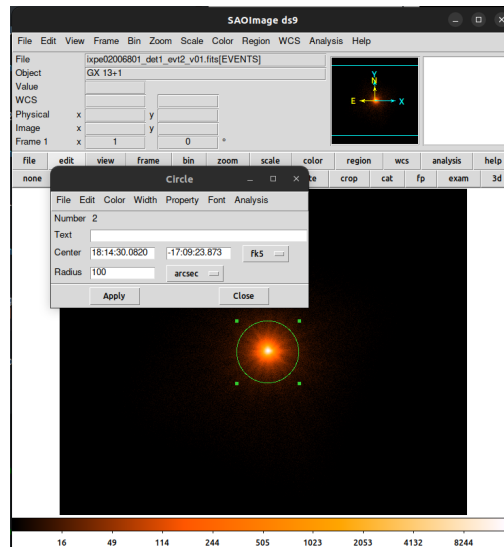
- Following the DS9 steps from [Sep. 23, 2025 \(Tuesday\)](#), on page 4, we obtain an enhancement of the image as follows:



- The source region is obtained by clicking on the centroid position and dragging to obtain a circle enclosing the source as follows:



- Double-clicking on the circular region opens the regions dialog. We have to ensure that the Center is in fk5 and the radius is in arcsec. Then we type 100 in the radius textbox to ensure that the source region inside is of radius 100". Then click Apply, followed by Close.



- So we have to create a source region file named `src.reg` in the `event_l2/` folder. The file `src.reg` has the following contents copied and pasted from the regions dialog:

```
circle(18:14:30.0820,-17:09:23.873,100.0")
```

- GX13+1 is a bright source and so, as per section 2 of [Di Marco et al. \(2023\)](#), we don't need to subtract any background photons.

May. 31, 2026 — Sunday

Reducing IXPE science data for analysis

- Obtaining the attitude files
 1. Go to the `hk/` folder and copy-paste the following attitude files into the `event_l2/` folder
`ixpe*_det?_att_v01.fits`
- Selecting the source and background regions
 1. Go to the `event_l2/` folder and open any of the L2 event files using DS9
`ixpe*_det?_evt2_v01.fits`
 2. Draw a circular source region with centroid at the location of the source and with radius in arcsec as per literature.

3. Create a region file, such as `src.reg` and type the region info with coordinates in `fk5` format and radius in arcseconds.
- Extracting the source and background photons
 1. We need to use `xselect` for this task.
 2. Make use of the steps mentioned in the journal entry dated **23 Sep. 2025 (Tuesday)**.
 3. This would create the I , Q and U spectra files for all 3 detectors, i.e. a total of 9 files.
 - Obtain the relevant RMF file
 1. Obtain the observation using the HEASARC search form.
 2. Locate the row containing the ObsID of interest.
 3. Copy the entry under the `time` column for that row.
 4. Ensure that the CALDB variable points to the calibration database with the following command on the Terminal:


```
echo $CALDB
```

if everything is right, then we will get an output such as the following:

```
https://heasarc.gsfc.nasa.gov/FTP/caldb
```

which is the remote CALDB host at HEASARC.
 5. If the above does not appear, then set the CALDB variable with the following command:


```
export CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb
```

or add the above line in the `.bashrc` file and compile it.
 6. Use the time value obtained in step 3 above in the `quzCIF` command. An example is as follows


```
quzCIF IXPE GPD DU1 - MATRIX 2023-10-17 16:41:45.850 -
```

where `GPD DU1` implies detector unit 1 of the gas pixel detector instrument. For detectors 2 and 3, we have to write `DU2` and `DU3` respectively, everything else remaining the same.
 7. After `quzCIF` has completed its run, it would give a list of FTP locations of 2 files.

8. Download the file whose name contains the string 'alpha075' using the `wget` command as follows:

```
wget -v <<FTP URL>>
```

For example:

```
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/ixpe/  
gpd/cpf/rmf/ixpe_d3_20211209_alpha075_01.rmf
```

9. Create soft links to these downloaded files, so that it is easier to refer to them, as follows

```
ln -s <<file name>> <<link name>>
```

For example:

```
ln -s ixpe_d2_20211209_alpha075_01.rmf gx13_d02_rmf.fits
```

following which we can use 'gx13_d02_rmf.fits' as an alias for 'ixpe_d2_20211209_alpha075_01.rmf'.

- Generate the ARF/MRF files

1. We need to use `ixpealcarf` for this task.
2. A helpful trick is to list the relevant event file and attitude files first using the following commands. For example, in case of detector 1:

```
ls det1_evt2  
ls det1_att
```

3. Then run the `ixpealcarf` as per steps mentioned in journal entry dated **16 Oct. 2025 (Thursday)**.

- Group the spectrum files

1. We need to use `fthedit` for this task.
2. Run the `fthedit` as per steps mentioned in journal entry dated **16 Oct. 2025 (Thursday)**.

June

Jun. 09, 2026 — Tuesday

- Found two nearby observations of GX 13+1 by IXPE and NuSTAR on the same day, i.e. 2024-02-25, as shown below

IXPE Master Catalog (ixmaster) Bulletin

Search radius used: 10.00'

Select	Services	obsid	name	ra	dec	time	exposure 1	exposure 2	exposure 3	processing date	public date	obs type	issue flag	status	Search Offset
☐ All															
☐ O R N S D B		04002901	GX 13+1	18 14 31.6	-17 09 27	2025-09-08 01:47:59.800	147888.5019	148058.7024	148092.6483	2026-03-27 16:06:33.388	2025-09-18	proposed	0	archived	0.091 (GX 13+1)
☐ O R N S D B		03003401	GX 13+1	18 14 31.6	-17 09 27	2024-04-20 20:06:58.889	99886.2991	100011.1685	100112.6865	2024-04-29 18:25:24.441	2024-05-01	proposed	0	archived	0.091 (GX 13+1)
☐ O R N S D B		02006801	GX 13+1	18 14 31.4	-17 09 25	2023-10-17 16:41:45.850	98203.6295	98298.5532	98299.4520	2023-10-23 20:10:10.177	2023-10-31	proposed	0	archived	0.054 (GX 13+1)
☒ O R N S D B		03001101	GX 13+1	18 14 31.6	-17 09 27	2024-02-25 12:55:38.500	90670.4378	90818.5112	90799.4449	2024-02-29 21:39:02.413	2024-03-05	proposed	0	archived	0.091 (GX 13+1)

4 rows retrieved from ixmaster

NuSTAR Master Catalog (numaster) Bulletin

Search radius used: 12.00'

Select	Services	name	ra	dec	time	obsid	status	exposure	observation mode	obs type	processing date	public date	issue flag	Search Offset
☐ All														
☒ O R N S D B		GX_13p1	18 14 41.5	-17 06 51	2024-02-25 12:56:10	30901010002	archived	32178	SCIENCE	N09	2024-02-27 02:24:00	2024-04-22	1	3.514 (GX 13+1)
☐ O R N S D B		GX_13p1	18 14 37.2	-17 08 12	2017-04-08 03:26:12	30301003002	archived	22711	SCIENCE	X16	2020-09-02 21:36:00	2018-05-08	0	1.845 (GX 13+1)
☐ O R N S D B		GX13p1	18 14 20.2	-17 11 29	2025-08-30 12:31:06	91101330002	archived	20544	SCIENCE	TOO	2025-09-01 02:24:00	2025-09-05		3.384 (GX 13+1)
☐ O R N S D B		GX_13p1	18 14 22.1	-17 11 46	2025-09-09 16:16:11	91101332002	archived	10579	SCIENCE	TOO	2025-09-11 12:00:00	2025-09-12	1	3.231 (GX 13+1)

4 rows retrieved from numaster

- Downloaded data for both observations.
- IXPE observation contained the same files and folders as before:
/auxil /event_l1 /event_l2 /hk
- NuSTAR observation contained the following files and folders:
/auxil /event_cl /event_uf /hk nu30901010002.cat.gz pipe.log

Jun. 15, 2026 — Monday

- Started the analysis of the 2024-02-25 data from IXPE.
- Providing a walkthrough of the steps below.

Background rejection filter

In Section 2.1 (IXPE), Bobrikova *et al.* (2024) state that they adopted the weighted analysis for their spectro-polarimetric study, following the baseline established by Di Marco *et al.* (2022a). This weighted scheme (using $w_k = \alpha^{0.75}$) is designed to improve polarimetric sensitivity by prioritizing well-reconstructed photoelectron tracks.

Bobrikova *et al.* (2024) also notes that they followed the recommendations of Di Marco *et al.* (2023b) regarding background handling. While they did not subtract the

background due to the source's brightness, they did utilize the background rejection strategy described in that source to improve the quality of their data.

As per [The Hitchhiker's Guide to IXPE Data Analysis](#) and [IXPE Quick Start Data Analysis Guide](#) both, we can know if the background-rejection algorithm was applied in the pipeline by checking if the FITS header keyword XPFLGBGD=T has been included in the EVENTS extension of the L2 event file. We can do this in two different ways:

1. Using the `fkeyprint` command in the Terminal, for example as
`fkeyprint ./event_l2/ixpe03001101_det1_evt2_v01.fits XPFLGBGD`
The output did not return XPFLGBGD=T and hence the background rejection algorithm was not applied.
2. Using the `fv` command, we can click on the Header buttons, then manually search for XPFLGBGD=T in the Primary, EVENTS and GTI extension headers.

Hence, we have to manually perform the background rejection. Given below are the steps to do so:

- The Python script named `filter_background.py` is available on the [GitHub page of IXPE-background](#).
- I have downloaded `filter_background.py` and placed it locally at `~/AstroDA/xray-pol/scripts/`
- Then I ran the script as follows:

```
python3 ~/AstroDA/xray-pol/scripts/filter_background.py
./event_l2/ixpe03001101_det1_evt2_v01.fits
./event_l1/ixpe03001101_det1_evt1_v01.fits --output rej
```

- I repeated the above step for detectors 2 and 3.
- This produced three background-rejected event files in the `/event_l2` folder with `_rej` appended to the existing filename.

Note: Other scripts can be obtained from the [Contributed IXPE Software](#) page.

Defining selection region

[Bobrikova et al. \(2024\)](#) used a circular region of radius 100'' with GX 13+1 at the centroid position. To do so in a similar manner here, we need to open the filtered `_rej` event files on DS9.

Problem: However, when I ran the command
`ds9 ./event_l2/ixpe03001101_det1_evt2_v01_rej.fits` & I encountered the following error:

```
Error in startup script: can't find package base64
while executing
"package require base64"
(file "/usr/share/saods9/library/ds9.tcl" line 216)
```

Fix: We can bypass this issue by telling the Terminal to ignore HEASoft's library paths just for the brief moment it launches DS9. To do so, I modified the above command as follows:
`env -u LD_LIBRARY_PATH ds9 ./event_l2/ixpe03001101_det1_evt2_v01_rej.fits` &

Data files for analysis

The *I*, *Q* and *U* PHA files were extracted using Xselect as described in the journal entry dated [23 Sep. 2025 \(Tuesday\)](#).

The ARF/MRF and RMF files were created using the `ixpecalcarf` and `quzCIF` tasks respectively.

Jun. 16, 2026 — Tuesday

- Continuation of the walkthrough follows.

Association of file headers

The various files were renamed using a simpler convention as follows:

- The source spectra files are renamed as, for example for detector 1:
`d1_I.src d1_Q.src d1_U.src`
- The background spectra files are renamed as, for example for detector 1:
`d1_I.bkg d1_Q.bkg d1_U.bkg`
- The ARF file is renamed as, for example for detector 1: `d1.arf`
- The MRF file is renamed as, for example for detector 1: `d1.mrf`
- A soft link is created for the appropriate RMF file as, for example for detector 1:
`d1.rmf -> ixpe_d1_20211209_alpha075_01.rmf`